MTH 201 Trig Substitutions & Partial Fractions

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Evaluate the integral $\int \sqrt{4-x^2}\,dx$ using a trigonometric substitution.

2. Instead of using a trigonometric substitution, evaluate the integral $\int_{-2}^{2} \sqrt{4-x^2}\,dx$ using simple geometry.

3. Suppose that $f(x)=\int \frac{4x}{x^2-2x-3}\,dx$ and $f(0)=15\ln(3)$. What is the value of $f(2)$?

4. Evaluate the integral $\int \frac{x^3+1}{x^2-3x+2}\,dx$.

5. Evaluate the integral $\int \frac{4x+2}{x^3-x^2+x-1}\,dx$

6. Evaluate the integral $\int \frac{e^x}{1-e^{2x}}\,dx$. (*Hint: Try the substitution* $u=e^x$)

7. Consider the function $f(x)=\frac{1}{x^n(x-a)}$ where $a\neq 0$ and n is a positive integer.

a) Setup up what the partial fraction decomposition of $f(x)$ look like. You do not need to find all numerators.

b) Determine the numerator for fraction with $(x-a)$ as its denominator.

8. The Weierstrass substitution is a technique for transforming a rational function of trigonometric functions into a rational function of *t*. It is named for the German mathematician, Karl Weierstrass, who is often called *the father of modern analysis*. Below is the substitution:

$$t = \tan\left(\frac{x}{2}\right), \text{ where } -\pi < x < \pi$$

a) Use a geometrical or analytic argument to show that under this transformation. (Hint: use some double angle identities on ii & iii)

i) $dx = \dfrac{2}{1+t^2}dt$

ii) $\sin(x) = \dfrac{2t}{1+t^2}$

iii) $\cos(x) = \dfrac{1-t^2}{1+t^2}$

b) Use this technique to evaluate the following integrals.

i) $$\int \frac{1}{1-\sin(x)+\cos(x)}\,dx$$

ii) $$\int \frac{1}{\tan(x)-\sec(x)}\,dx$$